MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

1/ TYPE DOD-003

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Reflex, intergral cavity, micrometer tuner, 8,100 to 12,400 MHz, waveguide output

ABSOLUTE RATINGS:

Parameter:	Ef	Ers	Er	F	Body T
Unit:	V	Vdc	Vdc	MHz	°C
Maximum:	6.9	550	-1,000	12,000	200
Minimum:	5.7	---	-20	8,100	---

PHYSICAL CHARACTERISTICS:

Mounting position:	Any	RF connection:	Waveguide, mates with UG - 39/U (MIL-F-3922/53)
Cooling:	Forced air (see note 1)		
Cathode:	Coated unipotential	Base:	Special pee wee, 3 pin (A3-1 pin reference only)
Dimensions:	See figure 1		
Cap:	Miniature (C1-4)		

TEST CONDITIONS:

Parameter:	Ef	Ers	Er	F	Body T	VSWR
Unit:	V	Vdc	Vdc	MHz	°C	σ'
	6.3	500	-200 to -400	F1	160 (max)	1.1:1 (max)

FREQUENCY		
F	MHz	Tolerance
F1	8,100	1%
F2	12,400	1%

GENERAL:

Qualification - required

1/ Formerly MXK-26/X-13

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
4027	Temperature coefficient	2	F = F1; Er/max Po	ΔF/ΔT	---	±0.25	MHz/°C
1031	Nonoperation vibration	-	t = 120 minutes; 40 minutes in each of three planes x, y, and z	---	---	---	---
	Quality conformance inspection, part 1	3					
4214	Cathode emission	4	Ef = 5.7 V	Δ Ik/Ik	---	-15	%
1256	Electrode current (resonator)	4		Irs	30	60	mAdc
4229	Total reflector current	4	Er = -60 to -300 Vdc	Ir	0	5	µAdc
4223	Mechanical tuning range	-	Er/max Po	F	8,100	12,400	MHz
4250	Power output	5	F = F1 and F2; Er/max Po	Po	100	---	mW
	Quality conformance inspection, part 2						
4289	Heater current (nonoperating)	-	Ef = 6.3 V	If	0.3	0.5	A
	Quality conformance inspection, part 3						
---	Life test	-	Group D; F = F1; Er/max Po	t	1,000	---	hrs
---	Life-test end point:						
4250	Power output	-	F = F1; Er/max Po	Po	75	---	mW

NOTES:

1. Forced-air cooling shall be used for resonator power inputs exceeding 10 watts. For maximum tube life, the tube temperature should be less than 100°C.
2. The initial frequency shall be allowed to stabilize by operation at practically constant ambient temperature for at least 10 minutes. The temperature range for the test shall be over the range of -10° to +40°C. The rate of change of ambient temperature shall not be greater than 2°C per minute.
3. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.
4. The tube shall not be oscillating during this test.
5. The power output shall be measured with the tube rigidly connected to a UG-39/U (MIL-F-3922/53) flange on an appropriate impedance transformer. The impedance transformer shall be adjusted for maximum power output.
6. Changes from previous issue. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DSA - ES

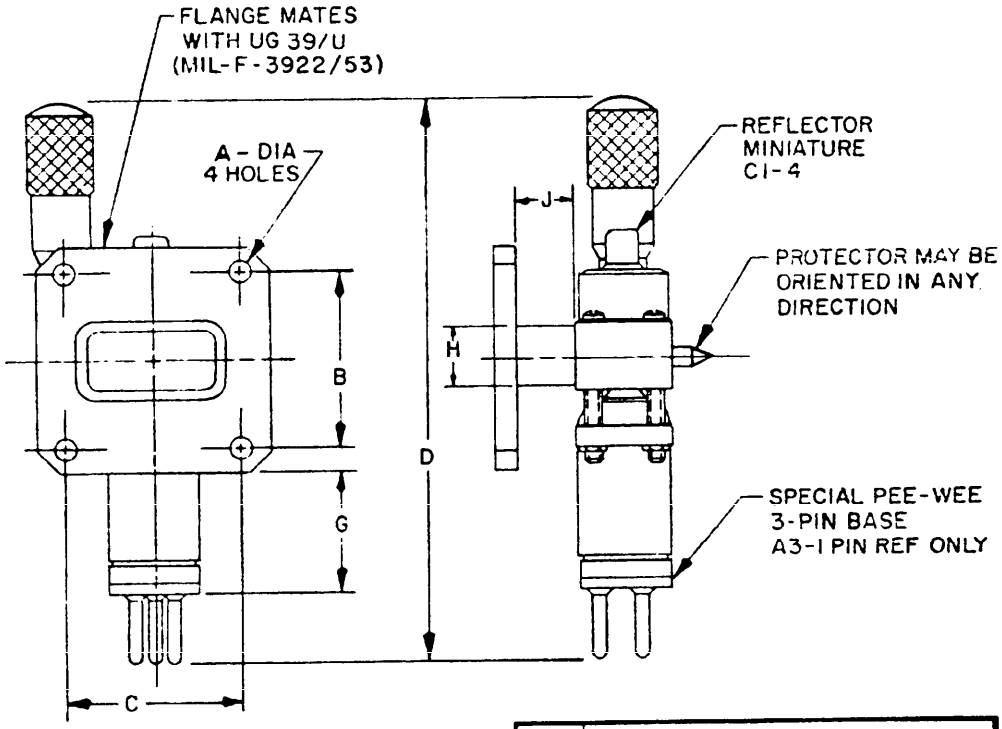
User activities:
Army - ME, WC
Navy - AS, OS
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3105)

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PIN CONNECTIONS

1	h	int con
2	k	
3	h	

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Ⓐ Qualification inspection		
E		1.875 (47.63)
G		.875 (22.23)
Quality conformance inspection, part 1		
D		4.500 (114.30)
Quality conformance inspection, part 2		
B	1.276 (32.41)	1.284 (32.61)
C	1.216 (30.89)	1.224 (31.09)
F		1.875 (47.63)
Nominal dimensions		
A	.185 (4.70)	
H	.425 (10.80)	
J	.421 (10.69)	
K	.915 (23.24)	

FIGURE 1. Outline drawing of electron tube type DOD-003.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED S/N 0102-014-1802

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